Artificial Intelligence (CSC-321)	**Instructor: Noreen Khalid**
Assignment - 3 Spring 2025	**Date assigned: 17th April**
Date Due: 22nd April	**Marks : 10**

Problem Statement and guidelines/Instructions:

- Solve all numerical questions with step-by-step calculations.
- Provide clear explanations for conceptual questions.
- Submit your assignment in Hard format & soft both.

Topics: (Decision Tree, Naïve Bayes, SVM & CNN).

[Mapped with Clo-01 & Clo-02]

1. Decision Tree

Conceptual Questions:

a. Explain how entropy and information gain are used in building a decision tree.

b. What is overfitting in decision trees? How can it be prevented?

c. Differentiate between Gini Index and Entropy. When would you prefer one over the other?

d. Given a dataset with binary classification and the following counts:

Attribute A	Yes	No
High	4	1
Medium	3	2
Low	2	3

Calculate:

- Entropy before the split.
- Information gain after splitting on Attribute A.

e. Given a dataset with attributes and class labels, simulate the first two levels of decision tree construction using the ID3 algorithm.

2. Support Vector Learning (SVL) / Support Vector Machines (SVM) Conceptual Questions:

a. What is the role of the kernel function in SVM?
b. How does soft margin SVM differ from hard margin SVM?
c. Explain the concept of support vectors and their importance.

Mathematical Problems:

4. Given two classes of data points in 2D:
   - Class A: (1,2), (2,3)
   - Class B: (3,3), (4,4)

   Construct a linear SVM decision boundary manually.
5. Given a quadratic kernel $K(x, y) = (x^T y + 1)^2$, compute the kernel matrix for the points $x_1 = (1, 0)$, $x_2 = (0, 1)$, $x_3 = (1, 1)$.

3. Naive Bayes
Conceptual Questions:

a. State and explain the Naive Bayes assumption. What are its pros and cons?
b. How does Naive Bayes handle continuous features?
c. Why is Naive Bayes considered a "generative" model?

Mathematical Problems:
d. Given the following:

Weather	Temperature	Play Tennis
Sunny	Hot	No
Overcast	Mild	Yes
Rainy	Cool	Yes
Sunny	Cool	Yes
Rainy	Hot	No

Use the Naive Bayes algorithm to classify the instance: (Sunny, Hot)

5. Calculate the posterior probability for a binary classification problem where:
   - Prior(P(Class=Yes)) = 0.6, Prior(P(Class=No)) = 0.4
   - Likelihoods: P(x1=1 | Yes)=0.5, P(x1=1 | No)=0.2
   - Given x1 = 1, compute the class probabilities.

4. Convolutional Neural Networks (CNNs)
Conceptual Questions:

a. What are the advantages of using CNNs for image processing tasks?
b. Explain the role of pooling layers in CNNs.
c. Describe how filters/kernels work in a convolution layer.

Mathematical Problems:

4. Given a 5×5 image and a 3×3 filter, compute the output of a convolution operation (stride = 1, no padding). Show the step-by-step matrix multiplication.
5. Consider a CNN with the following layers:
   - Input: 32×32×3
   - Conv layer: 5×5 filters, 6 filters, stride=1, padding=0
   - Pooling layer: 2×2 max-pooling, stride=2

   Calculate the output dimensions after each layer.

5. Neural network example with a comparable structure & dimensions, find parameter calculations.

Layer	Activation shape	Activation Size	# Parameters
Input:	(64, 64, 3)	12,288	0
CONV1 (f=3, s=1)	(62, 62, 16)	61,504	?
POOL1	(31, 31, 16)	15,376	0
CONV2 (f=3, s=1)	(29, 29, 32)	26,912	4,640
POOL2	(14, 14, 32)	6,272	0
FC1	(128, 1)	128	?
FC2	(64, 1)	64	8,256
Softmax	(5, 1)	5	325

Calculate the values of parameter CONV1 & FC1?